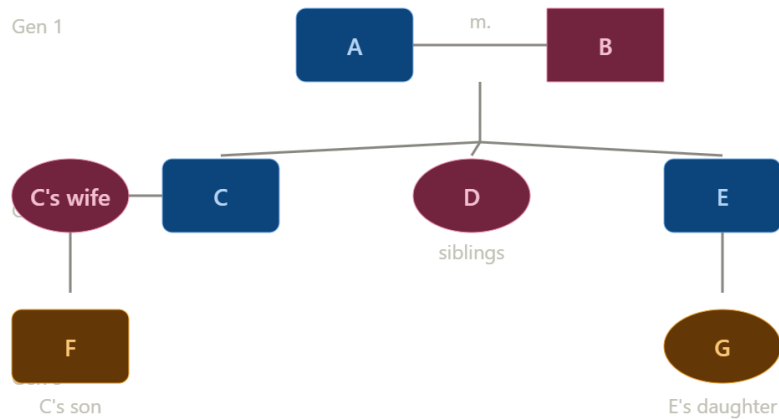


Blood Relation



F and G are cousins (same generation, different Gen-2 parents)

F is C's son → F calls D and E "uncle/aunt"; F calls A and B "grandparents"

Rule: same row = siblings/cousins/spouses · one row up = parent/uncle-aunt · one row down = child/niece-nephew

That tree shows the core method: **draw it, don't hold it in your head.** Squares = male, circles = female, horizontal line = marriage, vertical lines = parent→child. Every blood-relation puzzle collapses once it's on paper.

Core relation formulas (memorize these pairs):

Given	Relation
Father's/Mother's son	Brother (or self)
Father's/Mother's daughter	Sister (or self)
Father's father	Grandfather
Father's mother	Grandmother
Father's brother	Uncle (paternal)
Father's sister	Aunt (paternal)
Mother's brother	Uncle (maternal)
Mother's sister	Aunt (maternal)
Son's wife	Daughter-in-law

Given	Relation
Daughter's husband	Son-in-law
Husband's/wife's brother	Brother-in-law
Husband's/wife's sister	Sister-in-law
Sister's husband	Brother-in-law
Brother's wife	Sister-in-law
Uncle's/aunt's child	Cousin
Father's father's son (not father)	Uncle
Mother's father's son	Uncle (maternal)
Grandfather's only son	Father (if stated as only son)
Father's daughter's husband	Brother-in-law (or son-in-law of grandparent)

Generation-counting rule (the real shortcut):

- Same generation → siblings, cousins, or spouse
 - One generation up → parent, uncle, or aunt
 - One generation down → child, niece, or nephew
 - Two generations up/down → grandparent / grandchild
- Once you know how many rows apart two people are and whether the connecting person is male or female, the relation name is almost mechanical.

Coded/symbol-based blood relations (the trickiest exam variant — symbols replace words):

Typical coding scheme:

- $A + B \rightarrow A$ is father of B
- $A - B \rightarrow A$ is mother of B
- $A \times B \rightarrow A$ is brother of B
- $A \div B \rightarrow A$ is sister of B
- $A \times B$ combined with gender clues resolves brother vs. male cousin

Reading a chain: $P + Q - R$ means P is father of Q , and Q is mother of $R \rightarrow$ chain gives P is R 's grandfather (since Q is female, P is the paternal grandfather through a daughter — so technically P is R 's maternal grandfather, since Q , R 's mother, is P 's daughter). **This is**

exactly where students lose marks — always track gender at each link, don't just chain the words.

Method for any coded question:

1. Write the last statement first if the question asks about it, then work backward substituting.
2. Convert each symbol to a real relation word as you go.
3. Draw the mini family tree — even for coded questions, 3 nodes and 2 lines takes 10 seconds and prevents chain-reading errors.

Common placement question formats:

1. **Statement chain** — "Pointing to a photograph, X said, 'She is the daughter of my mother's only son.'" → decode step by step from the innermost clause outward.
2. **Symbol-coded chain** — as above, resolve symbol by symbol.
3. **Family tree with missing member** — table of 6-8 people, clues given, draw the tree once and answer all sub-questions from it.
4. **"How is X related to Y"** — the generation-distance rule solves this fastest once the tree is drawn.

Coding Decoding

Alphabet Number Line

A1 B2 C3 D4 E5 F6 G7 H8 I9 J10 K11 L12 M13 N14 O15 P16 Q17 R18 S19 T20 U21 V22 W23 X24 Y25 Z26

Alphabet Number Line

Z1 Y2 X3 W4 V5 U6 T7 S8 R9 Q10 P11 O12 N13 M14 L15 K16 J17 I18 H19 G20 F21 E22 D23 C24 B25 A26

EJOTY trick
E-5, J-10, O-15, T-20, Y-25 (multiples of 5)

opposite letter formula
opposite position = 27 - position

pairs: A-Z, B-Y, C-X, D-W, E-V, F-U, G-T, H-S, I-R, J-Q, K-P, L-O, M-N
e.g. opposite of H (pos 8) = 27-8 = 19 = S

	0	1	2	3	4
0	A	B	C	D	E
1	F	G	H	I	J
2	K	L	M2	M	O
3	P	Q	R	S	T
4	U				Y

Matrix coding grid pattern

reading the code

code = (row, column) or (column, row)

M sits at row 2, col 3

so M can be coded 23 or 32

check which order the given example uses, then apply that same order throughout

Both visuals above cover the two foundational tools — the alphabet number line (with EJOTY and opposite-letter shortcuts) and the row-column matrix grid. Here's the complete pattern set built on top of them:

Pattern 1 — Letter shifting / Caesar-style coding

Each letter is replaced by the letter a fixed number of steps ahead or behind in the alphabet.

- Rule: new position = (old position $\pm$ shift), wrapping around after Z back to A if needed.
- Example: shift +2 $\rightarrow$ CAT becomes EAV (C $\rightarrow$ E, A $\rightarrow$ C... wait, check direction consistently: A+2=C, T+2=V)
- Watch for **variable shift per letter** (e.g., 1st letter +1, 2nd letter +2, 3rd letter +3...) — a harder but very common placement variant.

Pattern 2 — Opposite/mirror letter coding

Uses the opposite-letter formula shown above: opposite position = 27 - position. Some papers code every letter as its mirror (A $\leftrightarrow$ Z, B $\leftrightarrow$ Y...) instead of shifting.

Pattern 3 — Number/position coding

Each letter is directly replaced by its forward or backward alphabet position ($A=1$ or $A=26$ depending on the scheme given). Always confirm from the example given in the question which direction is used before decoding the target word.

Pattern 4 — Matrix/coordinate coding

As shown in the grid: each letter's code is its (row, column) pair from a 5×5 or similar grid. The trickiest part is figuring out whether the code is written row-first or column-first — always deduce this from the solved example in the question stem, never assume.

Pattern 5 — Symbol/number substitution coding

Each letter or whole word is mapped to an arbitrary symbol or number with no numeric logic (pure lookup table) — e.g., "A is coded as 3, B is coded as 7..." These must be solved by direct substitution using the given key; there's no formula, just careful mapping and cross-checking for repeated letters.

Pattern 6 — Word/sentence substitution coding

"If 'apple' is called 'orange' and 'orange' is called 'mango'..." — a chain of swapped meanings. Method: build a substitution table first (don't try to resolve mid-sentence), then re-read the question using only the table.

Pattern 7 — Sum/difference coding

A word's code number is the sum (or product, or difference) of the alphabetical positions of its letters. Example: $CAT = C(3) + A(1) + T(20) = 24$. Always verify the operation (sum vs. product vs. average) against the given example — these questions almost always give one worked example to calibrate against.

Pattern 8 — Jumbled/reversed letter coding

The coded word is the original word's letters reversed, or reversed in groups (e.g., reverse each pair of letters). Test by reversing the given example first.

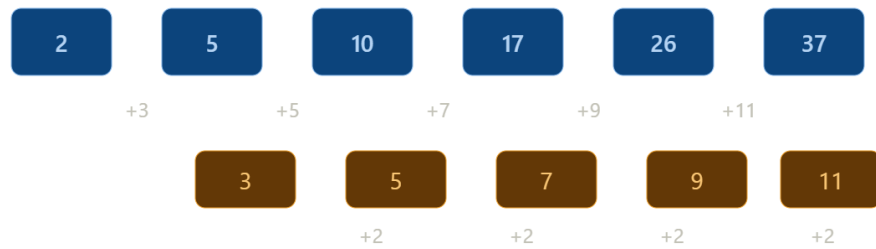
Universal solving method (works for every pattern above):

1. Never guess the rule from the target word — extract the rule from the **given example only**.
2. Apply the exact same operation, same direction, same order to the new word.
3. For coded sentences (word-for-word substitution across multiple given sentences), build a common word-to-code dictionary from all example sentences combined before answering — the answer word is often only derivable by combining two or three given sentences, not one.

Most common mistakes:

- Assuming forward alphabet order ($A=1$) when the question uses reverse order ($Z=1$) — always check with EJOTY or a known letter from the example.
- Missing that vowels/consonants might be coded with different rules within the same word.
- In matrix coding, transposing row and column by habit instead of checking the given example's order.

Number Series



second differences constant (+2) → this is a quadratic series, next term = $37 + 13 = 50$
rule: if 1st differences aren't constant, check 2nd differences before guessing multiplication

That ladder method (1st differences → 2nd differences) is the master key — it solves more series questions than any single formula. Here's the complete pattern catalogue:

- 1. Arithmetic series** — constant difference: $a_n = a_1 + (n-1)d$. Example: 4, 9, 14, 19... ($d=5$)
- 2. Geometric series** — constant ratio: $a_n = a_1 \times r^{n-1}$. Example: 3, 6, 12, 24... ($r=2$)
- 3. Quadratic/second-order series** — 1st differences form their own arithmetic series (as in the diagram above). Detect by taking differences of differences; if that row is constant, the series is quadratic.
- 4. Cubic/third-order series** — needs a third layer of differencing to become constant. Rare but appears in harder papers (2nd differences themselves form an AP).
- 5. Square-based series** — terms are squares $\pm$ a constant. Example: 3, 8, 15, 24, 35 → each is $n^2 + 2n$ → check by testing ($\sqrt{\text{term}}$) closeness to integers, or spot n^2 , n^2+1 , n^2-1 patterns.
- 6. Cube-based series** — terms are cubes $\pm$ constant. Example: 9, 28, 65, 126 → n^3+1 pattern ($2^3+1=9$, $3^3+1=28$...).
- 7. Prime number series** — 2, 3, 5, 7, 11, 13... or primes with a twist (every alternate prime, primes ± 1 , etc.)
- 8. Alternating operation series** — operations alternate rather than repeat. Example: 2, 6, 5, 15, 14, 42 ($\times 3$, -1 , $\times 3$, -1 ...). Always write out the operation under each gap; don't assume one operation throughout.
- 9. Two-stage/interleaved series** — odd positions follow one rule, even positions follow another. Example: 3, 20, 5, 18, 7, 16 → odd positions (3,5,7...) increase by 2; even positions (20,18,16...) decrease by 2. Split into two sub-series before analyzing.
- 10. Multiplication/division series** — each term is previous $\times$ or $\div$ a changing factor. Example: 2, 6, 24, 120 ($\times 3$, $\times 4$, $\times 5$...factorial-style growth).
- 11. Fibonacci-style series** — each term is the sum of the previous two (or a variant: sum of previous two minus a constant). Example: 1, 1, 2, 3, 5, 8, 13...

12. Mixed digit-sum series — next term formed from sum/product of digits of current term rather than the term itself. Example: 12, 15, 21, 24, 30 (each += sum of its own digits: $12+1+2=15$, $15+1+5=21$...).

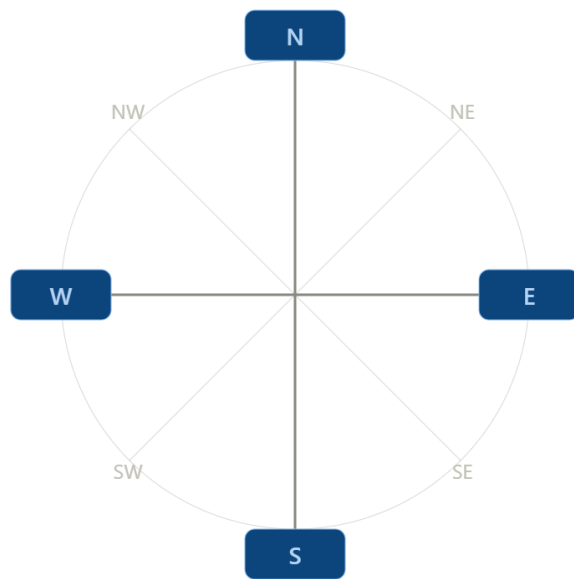
13. Wrong number / odd-one-out series — same pattern types as above, but one term is deliberately broken; find the rule from the majority of consistent gaps, then identify which single term violates it.

Universal solving checklist (apply in this exact order):

1. Check if it's a simple AP or GP first (constant difference or constant ratio) — fastest to rule out.
2. If not, take first differences. Are they constant? → arithmetic. Do they themselves form an AP/GP? → quadratic/exponential-differences.
3. If differences look irregular, check squares, cubes, or primes near each term.
4. If nothing fits a single rule, split into odd/even positions — check for a two-stage series.
5. If still nothing, check alternating operations (write the actual operation, not just the number, under each gap).
6. Last resort: digit-sum or digit-manipulation rule.

Biggest speed trap: committing to "it's $\times 2$ every time" after checking only the first two terms — always verify a hypothesized rule against **at least three consecutive gaps** before locking in an answer, since two-stage and alternating series often look identical to a simple series for the first couple of terms.

Direction Sense



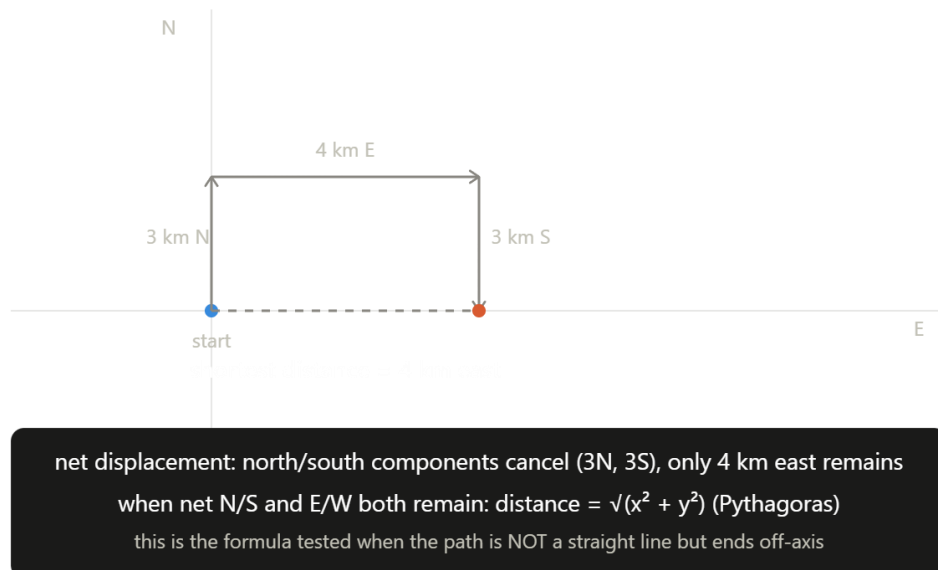
turn rules

facing N, left turn → W
facing N, right turn → E
facing E, left turn → N
facing E, right turn → S
180° turn → exact opposite
each 90° turn moves one quadrant
clockwise = right, anti-clockwise = left

shadow rule

sunrise: shadow falls west
sunset: shadow falls east
noon (tropics): shadow shortest
facing sun at sunrise → facing east

Now the method that solves 90% of direction problems — the coordinate/vector approach:



The one method to actually learn: coordinate/vector tracking. Assign start = (0,0), treat North/South as the y-axis and East/West as the x-axis, add or subtract distance for every leg of the journey, then read off the final point. Everything else below is a shortcut layered on top of this.

Core formulas and rules:

Situation	Rule
Straight-line net movement (all N/S or all E/W)	Simply add/subtract distances along that one axis
Final position off both axes	Shortest distance = $\sqrt{x^2 + y^2}$ — Pythagoras
Movements form a right-angle path (e.g., walked E then N)	Direct hypotenuse = $\sqrt{(\text{leg1}^2 + \text{leg2}^2)}$
Opposite-direction legs (N then S, or E then W)	They cancel — subtract the smaller from the larger
180° turn	Face exact opposite direction
90° left turn	Anti-clockwise: N→W→S→E→N
90° right turn	Clockwise: N→E→S→W→N
"Turn and walk," repeated	Update both position (x,y) and facing direction after every turn — track both, not just one

Direction-word to angle mapping (needed for "at what angle" or bearing-style questions):

N=0°/360°, NE=45°, E=90°, SE=135°, S=180°, SW=225°, W=270°, NW=315° — clockwise from North.

Shadow/sun-based rules (shown in the compass diagram above):

- At sunrise, shadow falls toward the west; a person facing the sun faces east.
- At sunset, shadow falls toward the east; a person facing the sun faces west.
- At noon (in tropical latitudes), the shadow is shortest/negligible — these questions can't be solved from shadow length alone at that time.

Common placement traps:

1. **Left/right turn is relative to current facing, not to North.** If you're facing east and turn left, you now face north — not west. Students who anchor turns to the compass instead of to their current heading get this wrong constantly.
2. **"Facing towards" vs "walking towards"** — a question may say a person is *facing* a direction without having walked in it; don't assume movement happened.
3. **Final distance vs final direction are two different asks** — "how far" wants the Pythagorean magnitude; "in which direction" wants the compass direction from start to end point, which needs the angle, not just distance.

4. **Multiple people, relative positions** — when two people's paths are described, plot both on the same coordinate grid; comparison questions ("who is farther east") come straight off the plot.

Seating Arrangement

1. Mathematical Formulas (Permutations & Combinations)

These formulas apply to quantitative aptitude questions where you must calculate the total number of possible arrangements.

Linear Arrangements (Straight Line)

- **Basic Linear Seating:** The number of ways to seat n people in a row is $n!$.
- **Restricted Seats:** The number of ways to seat n people in r available seats is ${}^nP_r = \frac{n!}{(n-r)!}$.
- **Specific People Always Together:** The number of ways to seat n people where a specific group of r people must sit together is $(n - r + 1)! \times r!$.
- **Specific People Never Together:** The number of ways to seat n people where 2 specific people must *never* sit together is $n! - ((n - 1)! \times 2!)$.
- **Alternate Seating (Equal Groups):** The number of ways to seat m men and n women alternately in a row (where $m = n$) is $2 \times m! \times n!$.

Circular Arrangements

- **Basic Circular Seating:** The number of ways to seat n people around a circular table is $(n - 1)!$.
- **Specific People Always Together:** The number of ways to seat n people in a circle where a group of r people must sit together is $(n - r)! \times r!$.
- **Alternate Seating (Equal Groups):** The number of ways to seat m men and n women alternately in a circle (where $m = n$) is $(m - 1)! \times n!$.
- **Indistinguishable Directions:** If clockwise and anti-clockwise arrangements are considered identical (like beads on a necklace, rarely used for people), the formula is $\frac{(n-1)!}{2}$.

Clock & Calendar

1. CLOCKS: Quick Reference Guide

Speeds & Relative Motion

Component	Distance / 60 Mins	Speed per Minute
Minute Hand	360°	6°
Hour Hand	30°	0.5°
Relative Speed	Minute hand gains	5.5°

The Core Formulas

1. The Master Angle Formula

To find the exact angle (θ) between the hands at a given time ($H:M$):

$$\theta = |30H - 5.5M|$$

Note: If your result is greater than 180°, subtract it from 360° to find the interior angle.

2. Finding the Exact Minute for a Specific Angle

To find the exact minute (M) past hour (H) when the hands hit a specific angle (θ):

$$M = \frac{2}{11} \times (30H \pm \theta)$$

Note: Calculate using both the + and – signs to find both occurrences within that hour.

3. Faulty Clocks (Gain/Loss per Day)

If a clock's hands coincide every x minutes (instead of the standard $\frac{720}{11}$ minutes), calculate the minutes gained or lost per day:

$$\left(\frac{720}{11} - x\right) \times \frac{60 \times 24}{x}$$

Note: A positive result means the clock is fast. A negative result means it is slow.

Frequency of Special Alignments

Alignment	Angle	12 Hours	24 Hours	Missing Time Frame
Coincide (Overlap)	0°	11 times	22 times	Between 11:00 and 1:00

Alignment	Angle	12 Hours	24 Hours	Missing Time Frame
Opposite (Straight)	180°	11 times	22 times	Between 5:00 and 7:00
Right Angle	90°	22 times	44 times	Between 2:00-4:00 & 8:00-10:00

Image Shortcut Rules

Image Type	Formula / Shortcut
Mirror Image	Subtract given time from 11:60 (or 23:60 for times past 12:00).
Water Image	Subtract given time from 18:30 (If minutes > 30, subtract from 17:90).

2. CALENDARS: Quick Reference Guide

The Odd Days Framework

Caleendars are solved by finding the remainder when total days are divided by 7.

Time Period	Total Days	Odd Days (Remainder)
Ordinary Year	365 (52 weeks + 1)	1
Leap Year	366 (52 weeks + 2)	2

Master Formula: Finding the Day of the Week

To find the specific day of the week for any historical or future date, add the following elements together and divide by 7. The **remainder** matches the Day Code.

$$\text{Total} = \text{Date} + \text{Month Code} + \text{Year} + \left\lfloor \frac{\text{Year}}{4} \right\rfloor + \text{Century Code}$$

The Universal Codes (Memorize These)

Type	Codes
Month Codes	144 (Jan-Mar) - 025 (Apr-Jun) - 036 (Jul-Sep) - 146 (Oct-Dec)

Type	Codes
	<i>Subtract 1 from Jan & Feb during a Leap Year.</i>
Century Codes	1600s: 6 1700s: 4 1800s: 2 1900s: 0 2000s: 6
Day Codes	1: Sun 2: Mon 3: Tue 4: Wed 5: Thu 6: Fri 0: Sat

Year Repetition Shortcuts

To find when an entire year's calendar will repeat, divide the given year by 4 and check the remainder.

Remainder	Add to Current Year	Example
0 (Leap Year)	+ 28 years	2024 repeats in 2052
1	+ 6 years	2025 repeats in 2031
2	+ 11 years	2026 repeats in 2037
3	+ 11 years	2027 repeats in 2038

Inequality

1. LOGICAL INEQUALITIES: Quick Reference Guide

These questions require you to evaluate a chain of symbols (e.g., $P > Q \geq R = S$) to determine if a conclusion is true.

The Priority Rule (King-Soldier-Public)

When checking the relationship between two letters, the highest-priority sign present between them dictates the final answer.

Priority Level	Forward Signs	Backward Signs	Rule / Outcome
1 (Highest)	$>$	$<$	The King: Overrides all other signs in the chain.
2 (Middle)	$\geq$	$\leq$	The Soldier: Overrides $=$, but loses to $>$, $<$.
3 (Lowest)	$=$	$=$	The Public: Ignored unless it is the ONLY sign present.

The "Gate Closed" Rule (Broken Chain)

- **Trigger:** Whenever you see **opposite-facing signs** (e.g., $>$ and $<$, or $\geq$ and $\leq$) anywhere between two elements.
- **Result:** The chain is broken. **No Relation** can be deduced. Any single definite conclusion is immediately **False**.

The "Either/Or" Master Checklist

You must select "Either Conclusion I or II follows" ONLY when these specific conditions align perfectly.

Case 1: The Broken Chain (No Relation)

1. Both elements in the conclusions are identical (e.g., A and B).
2. Both conclusions are individually **False**.
3. The conclusions cover **all three possibilities**: $>$, $<$, and $=$.

(Example: I. $A > B$, II. $A \leq B \Rightarrow$ Either I or II)

Case 2: The Soldier Chain (Split Sign)

1. Both elements in the conclusions are identical.
2. Both conclusions are individually **False**.

- The actual relation is a Priority 2 sign (e.g., $A \geq B$), and the conclusions explicitly **split** that sign into its two parts.

(Example: I. $A > B$, II. $A = B \Rightarrow$ Either I or II)

2. ALGEBRAIC INEQUALITIES: Quick Reference Guide

These patterns appear in the Quantitative section. The biggest trap here is forgetting how negative numbers and absolute values affect the equation.

The Golden Rule of Sign Flipping

You **MUST** reverse the inequality sign ($>$ becomes $<$) in these two scenarios:

- Multiplying or dividing both sides by a **negative number**.

$$-x > 5 \Rightarrow x < -5$$

- Taking the **reciprocal** of both sides (when both sides have the same sign).

$$x > y \Rightarrow \frac{1}{x} < \frac{1}{y}$$

The Wavy Curve Shortcut (Quadratic Inequalities)

For factored quadratics like $(x - a)(x - b)$ where $a < b$.

Format	Immediate Answer
$(x - a)(x - b) < 0$	$a < x < b$ (Value is strictly between the roots)
$(x - a)(x - b) \leq 0$	$a \leq x \leq b$ (Inclusive)
$(x - a)(x - b) > 0$	$x < a$ OR $x > b$ (Value is outside the roots)

The AM-GM Trick (For "Minimum Value" Questions)

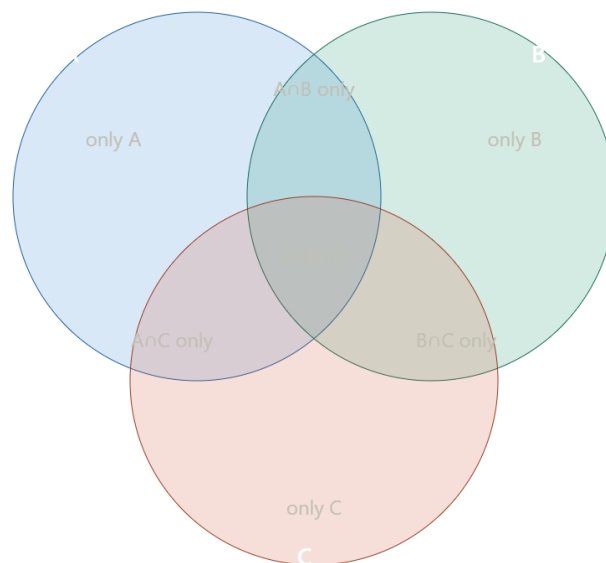
If a question asks for the minimum value of an expression with positive variables, apply the Arithmetic Mean $\geq$ Geometric Mean rule.

$$\frac{a + b}{2} \geq \sqrt{ab}$$

Classic Example: Find the minimum value of $x + \frac{1}{x}$ (for $x > 0$).

$$\frac{x + \frac{1}{x}}{2} \geq \sqrt{x \cdot \frac{1}{x}} \Rightarrow \text{Minimum value is 2.}$$

Venn Diagram



$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

That's the master 3-set map. Here are all the derived formulas built from those seven regions:

Two-set:

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- Only A = $n(A) - n(A \cap B)$; Only B = $n(B) - n(A \cap B)$
- Exactly one of A, B = $n(A) + n(B) - 2n(A \cap B)$
- Neither = Total - $n(A \cup B)$

Three-set (the ones exams actually test):

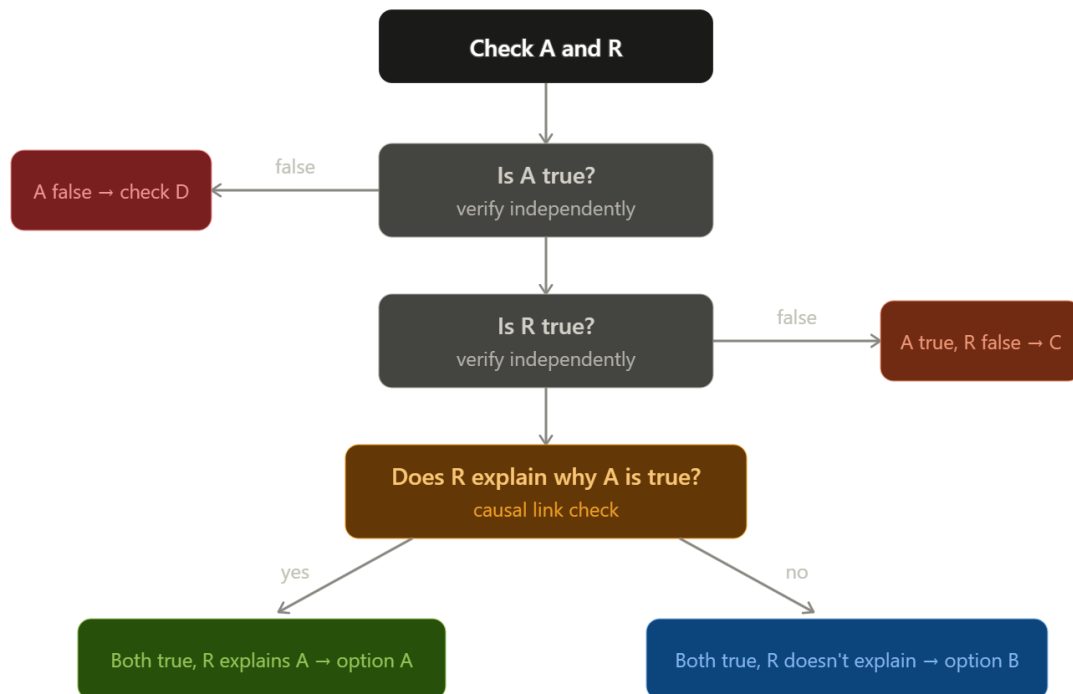
- Only A = $n(A) - n(A \cap B) - n(A \cap C) + n(A \cap B \cap C)$ — and similarly for only B, only C
- Exactly two of the three = $n(A \cap B) + n(B \cap C) + n(A \cap C) - 3 \cdot n(A \cap B \cap C)$
- Exactly one of the three = $n(A) + n(B) + n(C) - 2[n(A \cap B) + n(B \cap C) + n(A \cap C)] + 3n(A \cap B \cap C)$
- At least two = $n(A \cap B) + n(B \cap C) + n(A \cap C) - 2n(A \cap B \cap C)$
- Neither/none = Total - $n(A \cup B \cup C)$
- Complement: $n(A') = \text{Total} - n(A)$; De Morgan: $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$

The most common trap in placement papers: a question gives "plays cricket," "plays football," "plays hockey" totals plus pairwise overlaps and the all-three overlap, then asks for "exactly two sports" or "only one sport" — not the raw pairwise numbers. Always subtract the triple-overlap out of the pairwise numbers before answering "exactly two."

Second Venn pattern in reasoning sections — classification/relation diagrams (not numeric), e.g. "Doctors, Fathers, Men" — pick the figure showing the correct relationship:

Relation	Diagram shape
All A are B	Small circle A fully inside B
No A is B	Two separate, non-touching circles
Some A are B	Two circles partially overlapping
Some A are not B	Overlapping circles, one crescent implied outside
Independent/unrelated classes	Three separate circles

Now the assertion–reasoning decision logic:



The flowchart above is the whole game for A–R questions. A few things worth pinning down beyond the diagram:

Standard option key (memorize this exact wording — most placement papers use it verbatim):

- (A) Both A and R true, R is the correct explanation of A
- (B) Both A and R true, R is NOT the correct explanation of A
- (C) A true, R false
- (D) A false, R true
- (E) Both A and R false

Most frequent trap patterns:

1. **The "true but irrelevant" trap** — R states a fact that is factually correct but explains a *different* aspect of A, not the reason A happened. This is the classic (A)-vs-(B) confusion. Test it by asking: "If I only knew R, could I logically derive A from it?" If not, it's (B).
2. **The "sounds logical but is factually wrong" trap** — R reads like a textbook-perfect explanation, but contains a wrong fact (wrong number, wrong mechanism, reversed cause-effect). Evaluate R's truth *before* checking if it explains A — don't skip straight to plausibility.
3. **Reversed causality** — sometimes A and R are swapped in disguise (R is actually the effect of A, not the cause). This still scores as "R does not explain A" $\rightarrow$ (B), even though both statements are true.
4. **Partial truth** — if R is true only in some cases/contexts (not universally), treat it as false for this question unless the assertion specifies that context.